Lecture plan · for the lecturer

Linear programming and duality

Optimization · University · Lecture 1

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Formulate LPs.
- See why corners are optimal.
- Interpret duals.

Activity: Draw it, then price it

1. Warm-up (5 min): تفوّق على الموظفين
2. Draw (20 min): Pairs shade the feasible region and evaluate every vertex.
3. Dual (15 min): Write the dual and check both optimal values match.
4. Price (15 min): Explain each shadow price to the canteen manager in one sentence.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Which corner won?
- What does a zero shadow price mean?
- When does a price stop being valid?

Exit ticket

- Explain one shadow price in plain words.

Try it in your next lecture
Give pairs a two-variable canteen problem on graph paper; they shade the feasible region, test every corner and find the optimum, then interpret the solver’s shadow prices for the manager.

Seminar sheet · for students

Linear programming and duality

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Name _____ Class _____ Date _____

1 **Shadow price 5, extra hour costs 3. Buy it?**

◯ Yes ◯ No ◯ Shadow prices say nothing about this

2 **Solve at the corners.**

max 3x + 2y s.t. x + y ≤ 4, x + 3y ≤ 6, x, y ≥ 0. The optimum is _____ at x = _____.

Word bank: 12 · 11 · 3 · 4

3 **Which statements about LP duality are true? Choose all that apply.**

☐ Any feasible dual value bounds the primal optimum

☐ At optimality, primal and dual values are equal

☐ A non-binding constraint has a positive shadow price

☐ The dual has one variable per primal constraint

4 **A shadow price remains valid for any amount of relaxation of its constraint.**

◯ True ◯ False

5 **Match each term to its meaning.**

Vertex	• • • •	Value of one more unit of a resource
Weak duality	• • • •	Slack constraint means zero dual
Complementary slackness	• • • •	Dual values bound the primal
Shadow price	• • • •	Where an optimum can be found

✎ **Do your students spend more time pivoting tableaux or interpreting solutions? What would you rebalance?**